Bisection Method — Ultimate Cheat Sheet

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1. Method Setup

- Interval: $([a_0, b_0])$, where $(f(a_0)f(b_0) < 0)$.
- Iteration rule (midpoint):

$$x_n = \frac{a_n + b_n}{2}$$

- Interval update:
  - If $(f(x_n) = 0) \rightarrow$ root found.
  - If $(f(a_n)f(x_n) < 0) \rightarrow$ new interval $= ([a_n, x_n])$.
  - Else $\rightarrow$ new interval $= ([x_n, b_n])$.

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2. Error Formulas

(a) Interval Length Error (guaranteed bound)

$$E_n \leq \frac{b_n - a_n}{2} = \frac{b_0 - a_0}{2^n}$$

(b) Midpoint Absolute Error

$$E_n^{mid} = |x_n - x_{n-1}|$$

(c) Relative Midpoint Error

$$E_n^{rel} = \frac{|x_n - x_{n-1}|}{|x_n|}$$

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3. Iteration Count Formula

Using Interval Error:

$$n \geq \log_2 \left(\frac{b_0 - a_0}{\text{tol}} \right)$$

Alternate (sometimes seen):

$$n \geq \log_2 \left(\frac{b_0 - a_0}{2, \text{tol}} \right)$$

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4. Stopping Criteria (choose any)

- Interval error tolerance.
- Relative midpoint error tolerance.
- $|f(x_n)| \leq$ tolerance (function value small enough).

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5. Key Notes

- **Guaranteed convergence** if initial interval has sign change.
- Error shrinks by factor of 2 each iteration.
- Interval error = conservative (safe). Midpoint error = tighter (practical).
- Tolerance defines "good enough" closeness.

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Quick Workflow:

1. Pick ($[a_0, b_0]$).
2. Compute midpoint.
3. Check sign & shrink interval.
4. Track error.
5. Stop when error tol.

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